

ENTITY RELATIONSHIP (ER) DIAGRAM AND RELATIONAL SCHEMA

1. Objective

The selected application is a **Movie Booking System**. The database stores customer information, movie details, booking details, theatre information, showtime details, and seat information. The system allows customers to book tickets for movies and maintains information about the theatres where movies are shown. It also records the showtimes and seats associated with each booking. The design represents the relationships between customers, bookings, movies, theatres, showtimes, and seats.

2. Entities and Attributes

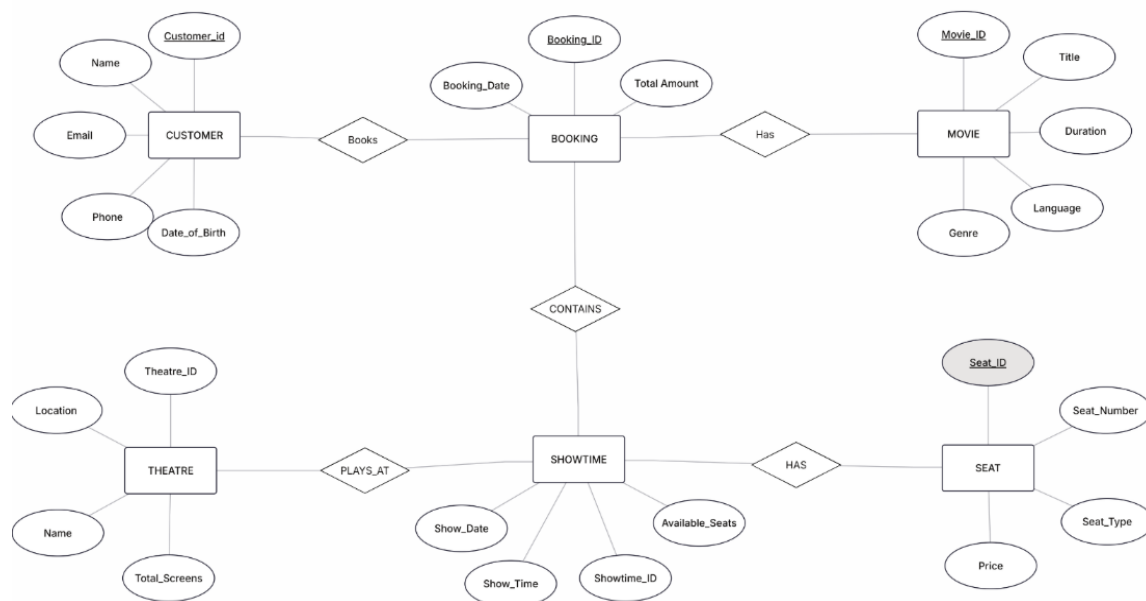
Entity	Primary Key	Other Attributes / Foreign Keys
CUSTOMER	customer_id (PK)	name, email, phone, date_of_birth
BOOKING	booking_id (PK)	booking_date, total_amount, customer_id (FK), movie_id (FK), showtime_id (FK)
MOVIE	movie_id (PK)	title, duration, language, genre
THEATRE	theatre_id (PK)	location, name, total_screens
SHOWTIME	showtime_id (PK)	show_date, show_time, available_seats, theatre_id (FK)
SEAT	seat_id (PK)	seat_number, seat_type, price, showtime_id (FK)

Entity Description

- **CUSTOMER** – Stores information about customers who book movie tickets.
- **BOOKING** – Stores booking details made by customers for movies.
- **MOVIE** – Stores information about movies, including title, duration, language, and genre.
- **THEATRE** – Stores information about theatres and their locations and screens.
- **SHOWTIME** – Stores the date and time at which a movie is scheduled to be shown.
- **SEAT** – Stores information about seats available for a particular showtime.

3. ER Diagram

The ER diagram below represents the entities, attributes, relationships, and overall structure of the **Movie Booking System**.



4. Relationships and Cardinalities

Entities	Relationship Cardinality Explanation		
CUSTOMER — BOOKING	Books	1 : M	One customer can make many bookings, while each booking belongs to one customer.
MOVIE — BOOKING	Has	1 : M	One movie can have many bookings, while each booking is associated with one movie.
THEATRE — SHOWTIME	Plays At	1 : M	One theatre can have multiple showtimes, while each showtime is associated with one theatre.
BOOKING — SHOWTIME	Contains	M : 1	Many bookings can be associated with a particular showtime, while each booking refers to one showtime.
SHOWTIME — SEAT	Has	1 : M	One showtime can have multiple seats, while each seat belongs to one showtime.

5. Relational Schema

The ER model is converted into relations using standard mapping rules. Entities become tables, primary keys uniquely identify records, and foreign keys represent relationships between tables.

Relation	Relational Schema
CUSTOMER	CUSTOMER(customer_id PK, name, email, phone, date_of_birth)
MOVIE	MOVIE(movie_id PK, title, duration, language, genre)
THEATRE	THEATRE(theatre_id PK, location, name, total_screens)
SHOWTIME	SHOWTIME(showtime_id PK, show_date, show_time, available_seats, theatre_id FK)
BOOKING	BOOKING(booking_id PK, booking_date, total_amount, customer_id FK, movie_id FK, showtime_id FK)
SEAT	SEAT(seat_id PK, seat_number, seat_type, price, showtime_id FK)

6. Primary Key and Foreign Key Identification

Table	Primary Key (PK)	Foreign Key(s) (FK)
CUSTOMER	customer_id	—
MOVIE	movie_id	—
THEATRE	theatre_id	—
SHOWTIME	showtime_id	theatre_id → THEATRE(theatre_id)
BOOKING	booking_id	customer_id → CUSTOMER(customer_id); movie_id → MOVIE(movie_id); showtime_id → SHOWTIME(showtime_id)
SEAT	seat_id	showtime_id → SHOWTIME(showtime_id)

7. Brief Relationship Explanation

- **Customer–Booking:** The relationship is **1:M** because one customer can make several bookings, while each booking is associated with one customer.
- **Movie–Booking:** The relationship is **1:M** because one movie can have multiple bookings, while each booking refers to one movie.
- **Theatre–Showtime:** The relationship is **1:M** because one theatre can have multiple showtimes, while each showtime is associated with one theatre.
- **Booking–Showtime:** The relationship is **M:1** because multiple bookings can be made for the same showtime, while each booking is associated with one showtime.
- **Showtime–Seat:** The relationship is **1:M** because one showtime can have multiple seats, while each seat belongs to one showtime.

8. ERD Reference

The ER diagram represents the entities, attributes, relationships, primary keys, and overall database structure of the Movie Booking System.

ERD Reference – Movie Booking System

